

Know-Surgery: A Unified Toolkit for Controllable Knowledge Update in Large Language Models

Anonymous ACL submission

Abstract

Deployed large language models (LLMs) require continuous knowledge updates to correct errors, remove harmful content, and incorporate new information. Existing tools address these needs in isolation: knowledge editing frameworks like EasyEdit focus solely on modifying factual associations, while unlearning toolkits like OpenUnlearning target only data removal. We present **Know-Surgery**, a unified toolkit that supports three complementary knowledge operations—*injection* (adding new knowledge via parameter-efficient fine-tuning), *deletion* (removing knowledge via machine unlearning), and *editing* (modifying specific knowledge via locate-and-edit methods)—within a single Hydra-configured framework. Know-Surgery integrates 48 methods across the three operations and their multimodal variants, covers five knowledge types (factual, event, conceptual, procedural, and rule-based), and extends to multimodal scenarios. Users can add new models, methods, and datasets in three steps through a registry-based architecture. Experiments on standard benchmarks (CounterFact, ZsRE, TOFU, MUSE) demonstrate that Know-Surgery reproduces published results while enabling direct cross-paradigm comparisons. The toolkit is available at <https://github.com/xxx/know-surgery> with a Flask + Vue web interface for interactive exploration.

1 Introduction

Large language models (LLMs) inevitably accumulate outdated facts, biased associations, and sensitive information during pre-training. Post-deployment, practitioners need to *update* this knowledge without costly full retraining. Three complementary paradigms have emerged to address different facets of this need:

- **Knowledge Editing** modifies specific factual associations through targeted parameter up-

dates (Meng et al., 2022, 2023; Mitchell et al., 2022a).

- **Machine Unlearning** removes the influence of particular training data to comply with privacy regulations or safety requirements (Maini et al., 2024; Shi et al., 2024).
- **Knowledge Injection** adds new capabilities or domain knowledge via parameter-efficient fine-tuning (Hu et al., 2022; Liu et al., 2024).

However, existing toolkits treat these paradigms in isolation. EasyEdit (Wang et al., 2024b) provides a comprehensive knowledge editing framework but does not support unlearning or injection. OpenUnlearning (Cho et al., 2025) unifies unlearning benchmarks but lacks editing capabilities. LLaMA-Factory (Zheng et al., 2024) focuses on fine-tuning without evaluating the impact on existing knowledge. This fragmentation forces researchers to maintain separate codebases, making cross-paradigm comparisons difficult.

We present **Know-Surgery**, a unified toolkit that integrates all three paradigms under a single framework. Our key contributions are:

1. A **unified architecture** based on Hydra configuration and a registry pattern, supporting 48 methods across editing (15), unlearning (14), and injection (6)—plus 13 multimodal variants—through a single entry point.
2. Coverage of **five knowledge types**—factual, event, conceptual, procedural, and rule-based—with 8+ editing datasets and standardized evaluation.
3. **Multimodal extensibility** through a dedicated sidecar supporting vision-language model unlearning on CLEAR and MLLMUBench benchmarks.
4. A **three-step extension protocol** enabling users to add new models, methods, or

081 datasets by implementing one class, registering
082 it, and creating a YAML config.

083 2 System Overview

084 Know-Surgery is built on a modular architecture
085 with four decoupled registries—MODEL, DATASET,
086 TRAINER, and EVALUATOR—orchestrated by Hydra
087 configuration. Figure 1 illustrates the overall de-
088 sign.

089 2.1 Unified Entry Point

090 All three operations share a single training script
091 (train.py) that dispatches to the appropriate
092 pipeline based on the mode parameter:

```
093 # Knowledge Editing  
094 python src/train.py mode=edit \  
095     trainer=edit/ROME model=Llama-3.1-8B \  
096     experiment=edit/zsre/default  
097  
098 # Machine Unlearning  
099 python src/train.py mode=unlearn \  
100     trainer=SimNPO model=Llama-3.1-8B \  
101     experiment=unlearn/tofu/default  
102  
103 # Knowledge Injection  
104 python src/train.py mode=inject \  
105     trainer=inject/LoRA model=Qwen2.5-7B \  
106     experiment=inject/alpaca/default  
107
```

109 2.2 Registry-Based Architecture

110 Each component type has a dedicated registry that
111 maps string names to Python classes:

- 112 • TRAINER_REGISTRY: 48 method implementa-
113 tions (15 edit + 14 unlearn + 6 inject + 13
114 multimodal)
- 115 • DATASET_REGISTRY: 20+ dataset handlers
116 covering QA, editing, injection, and multi-
117 modal formats
- 118 • MODEL_REGISTRY: supports any Hugging-
119 Face AutoModelForCausalLM-compatible
120 model via YAML config
- 121 • EVALUATOR_REGISTRY: 18 evaluators span-
122 ning TOFU, MUSE, WMDP, and edit-
123 specific metrics

124 Components are decoupled: methods do not de-
125 pend on specific datasets or models. A new com-
126 ponent requires only (1) a Python class inheriting
127 the base, (2) one registration line, and (3) a YAML
128 config file.

[Architecture diagram placeholder]
train.py → Registry dispatch → Edit/Unlearn/Inject
pipelines
Hydra configs combine Model + Trainer + Data + Eval

Figure 1: Know-Surgery architecture. A unified en-
try point dispatches to three pipelines through registry-
based component resolution.

2.3 Three Knowledge Operation Pipelines

Editing Pipeline. Edit requests pass through build_edit_requests → execute_edit_requests → save_edit_artifacts. Each editor implements an edit(requests) method that modifies model weights and returns a backup for rollback. Supported method families include locate-then-edit (ROME, MEMIT, AlphaEdit), meta-learning (MEND, MALMEN, InstructEdit), memory-based (GRACE, WISE, IKE, SERAC), and unstructured editing (UNKE, AnyEdit).

Unlearning Pipeline. Forget and retain datasets are combined into a ForgetRetainDataset that yields paired samples. Each unlearning trainer overrides compute_loss to implement its objective (gradient ascent, preference optimization, representation manipulation, etc.).

Injection Pipeline. PEFT adapters (LoRA, DoRA, AdaLoRA, LoReFT, BREP) are applied to the base model. The injection trainer handles adapter initialization, training, and optional merging.

3 Key Features

3.1 Comprehensive Method Library

Table 1 summarizes the methods integrated in Know-Surgery across three operation categories, totaling 35 text-based methods plus 13 multimodal methods.

3.2 Multi-type Knowledge Coverage

Unlike existing tools that focus primarily on factual triplets, Know-Surgery covers five knowledge types through 8+ editing datasets:

3.3 Multimodal Extension

Know-Surgery extends to multimodal scenarios through two pathways:

- **MM Unlearning** (7 methods): MMGradAscent, MMGradDiff, MMKLMin, MMNPO,

Op.	Category	Methods
Edit (15)	Locate-Edit	ROME, MEMIT, MEMIT_M, AlphaEdit
	Memory	GRACE, WISE, IKE, SERAC
	Meta-learn	MEND, MALMEN, InstructEdit
	Unified Unstruct.	UniKE, NMKE, UNKE, AnyEdit
Unl. (14)	Gradient	GradAscent, GradDiff, SGA
	Preference	NPO, DPO, SimNPO
	Represent.	RMU, FLAT, SOUL
	Others	UNDIAL, CEU, SatImp, WGA, PDU
Inj. (6)	PEFT	LoRA, DoRA, AdaLoRA, LoReFT, BREP
MM (13)	MM Unl.	MMGradAsc., MMNPO, MMVKD, ... (7)
	MM Edit	MM-IKE, MM-UniKE, ... (6)

Table 1: Methods supported in Know-Surgery. MEMIT_M = MEMIT_Merge.

Type	Datasets	Methods
Factual	CF, ZsRE, LEME	ROME, MEMIT, AlphaEdit
Event	ELKEN, AKEW	UnKE, MEMIT, AnyEdit
Concept	ConceptEdit	ROME, AlphaEdit
Proced.	EditEvery	AnyEdit, UNKE
Rule	RuleTaker, FOLIO	BREP, LoReFT

Table 2: Knowledge type coverage. CF=CounterFact.

MMRetainFT, MMUnlearner, MMVKD, evaluated on CLEAR and MLLMU-Bench.

- **MM Editing** (6 methods): MM-IKE, MM-GRACE, MM-WISE, MM-MEND, MM-SERAC, MM-UniKE, extending text editing methods to vision-language models (InternVL, LLaVA, Qwen-VL) via a shared MMEditMixin base class.

3.4 Three-Step Extension Protocol

Adding a new component requires only three steps:

1. **Implement:** Write a Python class inheriting the appropriate base class (EditTrainer, FinetuneTrainer, or InjectTrainer).
2. **Register:** Add one line to the registry (`_register_trainer(MyMethod)`).
3. **Configure:** Create a YAML file specifying the handler name and method-specific hyperparameters.

No changes to the core framework, data loading, or evaluation code are needed.

[WebUI screenshot placeholder]
Configuration panel + Results comparison view

Figure 2: Know-Surgery web interface for interactive experiment management.

4 Demonstration

Know-Surgery provides two interfaces for interactive exploration.

Command-Line Interface. Users run experiments with a single command, combining model, method, and dataset through Hydra overrides:

```
# Edit: ROME on CounterFact with Qwen2.5-7B
python src/train.py mode=edit \
  trainer=edit/ROME model=Qwen2.5-7B-Instruct \
  experiment=edit/counterfact/default

# Unlearn: SimNPO on TOFU
python src/train.py mode=unlearn \
  trainer=SimNPO model=Llama-3.2-1B-Instruct \
  experiment=unlearn/tofu/default \
  task_name=my_simnpo_run

# Inject: LoRA on Alpaca
python src/train.py mode=inject \
  trainer=inject/LoRA model=Qwen2.5-7B-Instruct \
  experiment=inject/alpaca/default

# Evaluate a saved checkpoint
python src/eval.py \
  experiment=eval/tofu/default \
  model.model_args.pretrained_model_name_or_path=
  saves/unlearn/my_simnpo_run
```

Web Interface. A Flask+Vue web UI (`new_ui/`) allows users to select models, methods, and datasets from dropdown menus, configure hyperparameters, launch experiments, and compare results visually. The frontend is built with Vite + TypeScript and communicates with a Python Flask backend via REST APIs.

Demo Video. A 2.5-minute screencast demonstrates the full workflow: selecting a method, configuring an experiment, running it, and analyzing results through both CLI and web interfaces.

5 Evaluation

We evaluate Know-Surgery on standard benchmarks to verify correctness and demonstrate cross-paradigm comparison capabilities.

5.1 Knowledge Editing

We compare representative editing methods on CounterFact and ZsRE. Metrics follow the stan-

Method	Rel.	Gen.	Loc.	Port.
<i>CounterFact</i>				
ROME	–	–	–	–
MEMIT	–	–	–	–
MEND	–	–	–	–
AlphaEdit	–	–	–	–
GRACE	–	–	–	–
IKE	–	–	–	–
<i>ZsRE</i>				
ROME	–	–	–	–
MEMIT	–	–	–	–
MEND	–	–	–	–
AlphaEdit	–	–	–	–
GRACE	–	–	–	–
IKE	–	–	–	–

Table 3: Knowledge editing results. Rel.=Reliability, Gen.=Generalization, Loc.=Locality, Port.=Portability. Values are percentages. *[TO BE FILLED after experiments]*

Method	Forget Quality	Model Utility
GradAscent	–	–
GradDiff	–	–
NPO	–	–
DPO	–	–
SimNPO	–	–
RMU	–	–
CEU	–	–
PDU	–	–
WGA	–	–

Table 4: Machine unlearning results on TOFU (forget10). *[TO BE FILLED]*

dard four-dimensional evaluation: Reliability (edit success rate), Generalization (rephrase accuracy), Locality (unrelated knowledge preservation), and Portability (reasoning transfer).

5.2 Machine Unlearning

We evaluate unlearning methods on the TOFU benchmark (forget10 split). Forget Quality measures how well the model forgets target data; Model Utility measures retained performance on non-target data.

5.3 Comparison with Existing Toolkits

6 Conclusion

We presented Know-Surgery, a unified toolkit for controllable knowledge update in large language models. By integrating knowledge editing, machine unlearning, and knowledge injection within a single registry-based framework, Know-Surgery enables researchers to directly compare and combine these complementary paradigms. The toolkit

Feature	Ours	EasyEdit	OpenUnl.
Edit methods	15	15+	0
Unlearn methods	14	0	11
Inject methods	6	1	5
Knowledge types	5	1–2	N/A
Multimodal	✓	✓	✓
Unified framework	✓	×	×
Config system	Hydra	Py API	Hydra
Web UI	✓	✓	×
Model configs	42	10+	10+

Table 5: Comparison with existing toolkits.

supports 35 text-based methods and 13 multi-modal methods across 42 model configurations, covers 5 knowledge types, and is easily extensible through a three-step protocol. Know-Surgery is open-source and includes both CLI and web interfaces for accessibility.

Limitations

Know-Surgery currently focuses on decoder-only language models. Support for encoder-decoder architectures (e.g., T5) is limited to methods that do not require specific layer path configurations. Evaluation of procedural and rule-based knowledge editing lacks established benchmarks, and our coverage in these areas relies on injection-based methods rather than direct editing.

Ethics Statement

Knowledge editing and unlearning tools carry inherent dual-use risks. While designed to improve model safety and factual accuracy, they could potentially be misused to inject misinformation or selectively remove safety-relevant knowledge. We encourage responsible use and recommend that practitioners document all knowledge modifications applied to deployed models.

References

- Ikhyun Cho, Woosung Koh, Kanghyun Ryu, Jiho Lee, and Jaeho Lee. 2025. OpenUnlearning: Accelerating LLM unlearning via unified benchmarking. *arXiv preprint*.
- Jingcheng Deng, Shumin Deng, Fangqian Wei, Kanyin Zheng, Ningyu Zhang, and Huajun Chen. 2024. UnKE: Unstructured knowledge editing in large language models. *arXiv preprint arXiv:2405.15349*.
- Yijiang River Dong, Hongzhou Lin, Mikhail Belkin, Ramon Huerta, and Ivan Vulić. 2025. UNDIAL: Self-distillation with adjusted logits for robust unlearning in large language models. In *Proceedings*

293	of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). ArXiv:2402.10052.	349
294		350
295		351
296	Taha Entesari, Arman Hatami, Rinat Khaziev, Anil Ramakrishna, and Mahyar Fazlyab. 2025. Constrained entropic unlearning: A primal-dual framework for large language models. <i>arXiv preprint arXiv:2506.05314</i> .	352
297		353
298		354
299		355
300		356
301	Chongyu Fan, Jiancheng Liu, Licong Lin, Jinghan Jia, Ruiqi Zhang, Song Mei, and Sijia Liu. 2024. Simplicity prevails: Rethinking negative preference optimization for LLM unlearning. <i>arXiv preprint</i> .	357
302		358
303		359
304		360
305	Junfeng Fang, Houcheng Jiang, Kun Wang, Yunshan Ma, Xiang Wang, Xiangnan He, and Tat-Seng Chua. 2024. AlphaEdit: Null-space constrained knowledge editing for language models. <i>arXiv preprint arXiv:2410.02355</i> .	361
306		362
307		363
308		364
309		365
310	Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. 2023. Aging with GRACE: Lifelong model editing with discrete key-value adaptors. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	366
311		367
312		368
313		369
314		370
315	Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In <i>International Conference on Learning Representations (ICLR)</i> .	371
316		372
317		373
318		374
319		375
320	Jiahao Huo, Yibo Yan, Xu Zheng, Yuanhuiyi Lyu, Xin Zou, Zhihua Wei, and Xuming Hu. 2025. MMUnlearner: Reformulating multimodal machine unlearning in the era of multimodal large language models. In <i>Findings of the Association for Computational Linguistics: ACL 2025</i> , pages 7190–7206. Association for Computational Linguistics.	376
321		377
322		378
323		379
324		380
325		381
326		382
327	Jinghan Jia, Yihua Zhang, Yimeng Zhang, Jiancheng Liu, Bharat Runwal, James Diffenderfer, Bhavya Kaikhura, and Sijia Liu. 2024. SOUL: Unlocking the power of second-order optimization for LLM unlearning. In <i>Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> . ArXiv:2404.18239.	383
328		384
329		385
330		386
331		387
332		388
333		389
334	Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berber, Alice Gatti, Justin D. Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, Gabriel Mukobi, Nathan Helm-Burger, Rassin Lababidi, Lennart Justen, Andrew B. Liu, Michael Chen, Isabelle Barrass, Oliver Zhang, Xiaoyuan Zhu, and 36 others. 2024. The WMDP benchmark: Measuring and reducing malicious use with unlearning. <i>arXiv preprint</i> .	390
335		391
336		392
337		393
338		394
339		395
340		396
341		397
342		398
343	Sirui Liang, Pengfei Cao, Jian Zhao, Cong Huang, Jun Zhao, and Kang Liu. 2025. Bias-restrained prefix representation finetuning for mathematical reasoning. <i>arXiv preprint arXiv:2511.10707</i> .	399
344		400
345		401
346		402
347	Jinzhe Liu, Junshu Sun, Shufan Shen, Chenxue Yang, and Shuhui Wang. 2025. Edit less, achieve	
348	more: Dynamic sparse neuron masking for life-long knowledge editing in LLMs. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> . ArXiv:2510.22139.	
	Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. 2024. DoRA: Weight-decomposed low-rank adaptation. <i>arXiv preprint</i> .	
	Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. 2024. TOFU: A task of fictitious unlearning for LLMs. In <i>International Conference on Machine Learning (ICML)</i> .	
	Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	
	Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2023. Mass-editing memory in a transformer. In <i>International Conference on Learning Representations (ICLR)</i> .	
	Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D. Manning. 2022a. Fast model editing at scale. In <i>International Conference on Learning Representations (ICLR)</i> .	
	Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D. Manning, and Chelsea Finn. 2022b. Memory-based model editing at scale. In <i>International Conference on Machine Learning (ICML)</i> .	
	Kaihang Pan, Zhaoyu Fan, Juncheng Li, Qifan Yu, Hao Fei, Siliang Tang, Richang Hong, Hanwang Zhang, and Qianru Sun. 2024. Towards unified multimodal editing with enhanced knowledge collaboration. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> . Spotlight. ArXiv:2409.19872.	
	Zirui Pang, Hao Zheng, Zhijie Deng, Ling Li, Zixin Zhong, and Jiaheng Wei. 2026. Label smoothing improves gradient ascent in LLM unlearning. In <i>International Conference on Learning Representations (ICLR)</i> . ArXiv:2510.22376.	
	Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	
	Weijia Shi, Jaechan Lee, Yangsibo Huang, Sadhika Malladi, Jieyu Zhao, Ari Holtzman, Daogao Liu, Luke Zettlemoyer, Noah A. Smith, and Chiyuan Zhang. 2024. MUSE: Machine unlearning six-way evaluation for language models. <i>arXiv preprint</i> .	
	Zhijie Tan, Hanyu Duan, Jiancheng Liu, and Weizhi Wang. 2024. MALMEN: Fast model editing at scale. In <i>International Conference on Learning Representations (ICLR)</i> . ArXiv:2304.00740.	

403	Bozhong Tian, Siyuan Cheng, Xiaozhuan Liang, Ningyu Zhang, Yi Hu, Kouying Xue, Yanjie Gou, Xi Chen, and Huajun Chen. 2024. InstructEdit: Instruction-based knowledge editing for large language models. <i>arXiv preprint</i> .	458
404		459
405		460
406		461
407		
408	Peng Wang, Zexi Li, Ningyu Zhang, Ziwen Xu, Yunzhi Yao, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. 2024a. WISE: Rethinking the knowledge memory for lifelong model editing of large language models. In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	462
409		463
410		464
411		465
412		466
413		
414	Peng Wang, Ningyu Zhang, Xin Xie, Yunzhi Yao, Bozhong Tian, Mengru Wang, Zekun Xi, Siyuan Cheng, Kangwei Liu, Guozhou Zheng, and Huajun Chen. 2024b. EasyEdit: An easy-to-use knowledge editing framework for large language models. In <i>Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)</i> .	467
415		468
416		469
417		470
418		
419		
420		
421	Qizhou Wang, Jin Peng Zhou, Zhanke Zhou, Saebyeol Shin, Bo Han, and Kilian Q. Weinberger. 2025a. Rethinking LLM unlearning objectives: A gradient perspective and go beyond. In <i>International Conference on Learning Representations (ICLR)</i> . G-effect / WGA. arXiv:2502.19301.	471
422		472
423		473
424		
425		
426		
427	Yaxuan Wang, Jiaheng Wei, Chris Yuhao Liu, Jinlong Pang, Quan Liu, Ankit Parag Shah, Yujia Bao, Yang Liu, and Wei Wei. 2025b. LLM unlearning via loss adjustment with only forget data. In <i>International Conference on Learning Representations (ICLR)</i> . FLAT. arXiv:2410.11143.	474
428		475
429		
430		
431		
432		
433	Yuhang Wang, Zhenxing Niu, Haoxuan Ji, Guangyu He, Haichang Gao, and Gang Hua. 2024c. Robust MLLM unlearning via visual knowledge distillation. <i>arXiv preprint arXiv:2512.11325</i> .	476
434		477
435		
436		
437	Zhengxuan Wu, Aryaman Arora, Zheng Wang, Atticus Geiger, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. 2024. LoReFT: Low-rank linear subspace ReFT. <i>arXiv preprint</i> .	478
438		479
439		480
440		
441	Bo Yang. 2025. CE-U: Cross entropy unlearning. <i>arXiv preprint arXiv:2503.01224</i> .	481
442		482
443	Puning Yang, Jianhao Huang, Jian Lou, Yinzhi Cao, and Jian Liu. 2025. Exploring criteria of loss reweighting to enhance LLM unlearning. In <i>International Conference on Machine Learning (ICML)</i> . SatImp. arXiv:2505.11953.	483
444		484
445		
446		
447		
448	Yuanshun Yao, Xiaojun Xu, and Yang Liu. 2023. Large language model unlearning. In <i>NeurIPS 2023 Workshop on Socially Responsible Language Modelling Research (SoLaR)</i> .	485
449		486
450		487
451		
452	Qingru Zhang, Minshuo Chen, Alexander Bukharin, Nikos Karampatziakis, Pengcheng He, Yu Cheng, Weizhu Chen, and Tuo Zhao. 2023. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning. In <i>International Conference on Learning Representations (ICLR)</i> .	488
453		489
454		
455		
456		
457		
	Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. 2024. Negative preference optimization: From catastrophic collapse to effective unlearning. <i>arXiv preprint</i> .	
	Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang. 2023. Can we edit factual knowledge by in-context learning? In <i>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> .	
	Yaowei Zheng, Richong Zhang, Junhao Zhang, Yanhan Ye, Zheyang Luo, Zhangchi Feng, and Yongqiang Ma. 2024. LlamaFactory: Unified efficient fine-tuning of 100+ language models. <i>arXiv preprint</i> .	
	A Complete Method Reference	
	B Dataset Summary	
	C Supported Models	
	Know-Surgery provides 42 pre-configured model YAML files covering:	
	• Llama family: Llama-2-7B/13B, Llama-3.1-8B, Llama-3.2-1B/3B	
	• Qwen family: Qwen2.5-0.5B/1.5B/3B/7B/14B-Instruct, Qwen2.5VL-7B, Qwen2VL-2B, Qwen3-VL-4B	
	• Others: Mistral-7B, Phi-3/3.5, Gemma-2B/7B/9B, Yi-1.5-6B/9B, GLM-4-9B, InternLM2.5-7B, DeepSeek-R1-Distill, GPT-2/J, Baichuan2-7B/13B	
	• Vision-Language: InternVL2.5-2B/8B, LLaVA-1.5-7B/13B, LLaVA-OneVision-7B, InstructBLIP-Vicuna-7B, BLIP2-OPT-2.7B	
	Any HuggingFace-compatible model can be added by creating a single YAML configuration file.	

Method	Category	Reference
<i>Editing (15)</i>		
ROME	Locate-Edit	Meng et al. (2022)
MEMIT	Locate-Edit	Meng et al. (2023)
MEMIT_Merge	Locate-Edit	Meng et al. (2023)
AlphaEdit	Locate-Edit	Fang et al. (2024)
MEND	Meta-learn	Mitchell et al. (2022a)
MALMEN	Meta-learn	Tan et al. (2024)
InstructEdit	Meta-learn	Tian et al. (2024)
GRACE	Memory	Hartvigsen et al. (2023)
WISE	Memory	Wang et al. (2024a)
IKE	Memory	Zheng et al. (2023)
SERAC	Memory	Mitchell et al. (2022b)
UniKE	Unified	Pan et al. (2024)
NMKE	Unified	Liu et al. (2025)
UNKE	Unstruct.	Deng et al. (2024)
AnyEdit	Multi-strat.	Deng et al. (2024) [†]
<i>Injection (6)</i>		
LoRA	PEFT	Hu et al. (2022)
DoRA	PEFT	Liu et al. (2024)
AdaLoRA	PEFT	Zhang et al. (2023)
LoReFT	PEFT	Wu et al. (2024)
BREP	PEFT	Liang et al. (2025)
<i>Unlearning (14)</i>		
GradAscent	Gradient	Yao et al. (2023)
GradDiff	Gradient	Maini et al. (2024)
SGA	Gradient	Pang et al. (2026)
NPO	Preference	Zhang et al. (2024)
SimNPO	Preference	Fan et al. (2024)
DPO	Preference	Rafailov et al. (2023)
RMU	Represent.	Li et al. (2024)
FLAT	Represent.	Wang et al. (2025b)
SOUL	Represent.	Jia et al. (2024)
UNDIAL	Other	Dong et al. (2025)
CEU	Other	Yang (2025)
SatImp	Other	Yang et al. (2025)
WGA	Other	Wang et al. (2025a)
PDU	Other	Entesari et al. (2025)
<i>Multimodal (13)</i>		
MMGradAscent	MM-Unl.	Yao et al. (2023)
MMGradDiff	MM-Unl.	Maini et al. (2024)
MMKLMin	MM-Unl.	–
MMNPO	MM-Unl.	Zhang et al. (2024)
MMRetainFT	MM-Unl.	–
MMUnlearner	MM-Unl.	Huo et al. (2025)
MMVKD	MM-Unl.	Wang et al. (2024c)
MM-IKE	MM-Edit	Zheng et al. (2023)
MM-GRACE	MM-Edit	Hartvigsen et al. (2023)
MM-WISE	MM-Edit	Wang et al. (2024a)
MM-MEND	MM-Edit	Mitchell et al. (2022a)
MM-SERAC	MM-Edit	Mitchell et al. (2022b)
MM-UniKE	MM-Edit	Pan et al. (2024)

Table 6: Complete method reference. 48 methods total: 15 text edit + 14 text unlearn + 6 inject (5 PEFT + full fine-tuning) + 7 MM unlearn + 6 MM edit. [†]This AnyEdit aggregates MEMIT, AlphaEdit, and UNKE strategies in a unified interface (distinct from the same-named ICML’25 method).

Dataset	Knowledge Type	Size
CounterFact	Factual	21,919
ZsRE	Factual	10,000
LEME	Factual (long-form)	7,390
ELKEN	Event	1,847
UnKE	Event/Temporal	1,000
AKEW	Event/Multi-hop	975+
ConceptEdit	Conceptual	452
EditEvery	Procedural	552
MQuAKE	Multi-hop	3,000+
TOFU	Unlearn	4,000
MUSE-News	Unlearn	6.5M tokens
MUSE-Books	Unlearn	6.5M tokens
WMDP	Unlearn (safety)	3,668
RWKU	Unlearn	200 entities
CLEAR	MM Unlearn	3,700
MLLMU-Bench	MM Unlearn	500+
MMEdit	MM Edit	1,000+
MMKE-Bench	MM Edit	2,000+
VLKEB	MM Edit	1,500+
Alpaca	Inject	52,000
ShareGPT	Inject	90,000+

Table 7: Datasets supported in Know-Surgery.